Wesley Wei Qian

wesley.chin0919@gmail.com
DrQ.ai · Scholar · GitHub

Experience

VP, Engineering & Research (Founding Team) | Osmo
Sep 2022 - Present

- I joined Osmo as its first research engineer and now lead engineering and research as VP while remaining hands-on in model development, scientific infrastructure, and experimental design. Our 10+ person team of software engineers, ML researchers, lab associates, and human-data experts takes AI-for-science from question to physical validation.
- **Computational & Experimental Science:** Partnered with perfumers, lab scientists, and annotators to turn ambiguous formulation questions into models and experiments. Shipped cost-and-safety optimization for our formula model, anchoring a multi-million-dollar CPG enterprise deal. Built a GNN that predicts intensity from molecular structure, lifting our ingredient-discovery hit rate. Worked with analytical chemists to "teleport" a physical plum scent, translating GC-MS signatures into digital profiles and validating the physical reconstruction end-to-end.
- **Agents for Scientific Work:** Architected Studio, an agentic platform that turns brand briefs into formulas and connects sales, manufacturing, and perfumery. Frontier LLMs, fine-tuned with SFT and RL, use tools exposed through MCP servers to access internal perfumery capabilities and chemical data. An active-learning loop turns perfumer feedback and failure cases into evaluations and training data.
- **Scientific Data & Infrastructure:** Built the post-Google ML stack from scratch (PyTorch, GCP, W&B), including data ingestion, workflow orchestration, experiment tracking, evaluation, and deployment. Unified molecular structures, fragrance materials, analytical instrument signals, and human-perception labels into interfaces used by researchers and AI agents.

Student Researcher | Google
May 2018 - Sep 2022

- **Olfactory ML:** Built receptor-binding and metabolic-activity models from molecular and biological data, helping catalyze Osmo's $60M spin-out. The work led to one utility patent and four publications featured in major news outlets.
- **Generative ML for Cell Imaging:** First-authored a generative approach to batch-effect correction in high-content cellular imaging, published in Bioinformatics and open-sourced as part of Google's TF-GAN library.
- **Genomics & Structural Variants:** Built ML filtering for structural variant calling and a ~100× faster read-alignment method, filed as a utility patent.

Intern | DeepMind
Sep 2021 - Dec 2021

- **Structural Bioinformatics:** Worked with the AlphaFold team to train and evaluate large-scale protein annotation models in JAX, transferring learned structure representations from AlphaFold2.

Software Engineering Intern | Uber
Summer 2016 & 2017

- **[2017] Sensor & Machine Learning:** Built a CRF variant to infer Uber Eats delivery events from mobile sensor data. Won 1st place at Uber's internal ML poster session.
- **[2016] Mobile Dev. & Engineering:** Built a web-based forensics tool for mobile UI testing that synchronized logs with video timestamps, cutting internal developer debugging time by ~50%.

Education

University of Illinois Urbana Champaign
2017 - 2022

Doctor of Philosophy in Computer Science
Dissertation: Machine learning for drug discovery and beyond (advisor: *Jian Peng*)
GPA: 4.00 / 4.00 | University Fellowship | Richard T. Cheng Endowed Fellowship

Brandeis University
2013 - 2017

Bachelor of Science in Computer Science and Neuroscience
GPA: 3.96 / 4.00 | Summa Cum Laude | Phi Beta Kappa (Junior) | Schiff Fellowship

Publications (*equal contribution)

- **Foundation models for discovery and exploration in chemical space**
  arXiv (2025)
  Alexius Wadell, Anoushka Bhutani, Victor Azumah, Austin R. Ellis-Mohr, Andrew J. Stier, Kareem Hegazy, Alexander Brace, Hancheng Zhao, Celia Kelly, Anuj K. Nayak, Yuhan Chen, Dimitrios Simatos, Hongyi Lin, Murali Emani, Venkatram Vishwanath, Kevin Gering, Melisa Alkan, Tom Gibbs, Jack Wells, **Wesley W. Qian**, Richard C. Gerkin, Benjamin Amorelli, Alexander B. Wiltschko, Lav R. Varshney, Bharath Ramsundar, Karthik Duraisamy, Michael W. Mahoney, Arvind Ramanathan, Venkatasubramanian Viswanathan
- **A deep learning and digital archaeology approach for mosquito repellent discovery**
  Chemical Senses (2025)
  Jennifer N. Wei*, Carlos Ruiz*, Marnix Vlot*, Benjamin Sanchez-Lengeling, Brian K. Lee, Luuk Berning, Martijn W. Vos, Rob WM Henderson, **Wesley W. Qian**, Jacob N. Sanders, D. Michael Ando, Kurt M. Groetsch, Richard C. Gerkin, Alexander B. Wiltschko, Jeffrey A. Riffell, Koen J. Dechering
- **New York Smells: a large multimodal dataset for olfaction**
  arXiv (2025)
  Ege Ozguroglu, Junbang Liang, Ruoshi Liu, Mia Chiquier, Michael DeTienne, **Wesley W. Qian**, Alexandra Horowitz, Andrew Owens, Carl Vondrick
- **Pervasive mislocalization of pathogenic coding variants underlying human disorders**
  Cell (2024)
  Jessica Lacoste, Marzieh Haghighi, Shahan Haider, Chloe Reno, Zhen-Yuan Lin, Dmitri Segal, **Wesley W. Qian**, Xueting Xiong, Tanisha Teelucksingh, Esteban Miglietta, Hamdah Shafqat-Abbasi, Pearl V. Ryder, Rebecca Senft, Beth A. Cimini, Ryan R. Murray, Chantal Nyirakanani, Tong Hao, Gregory G. McClain, Frederick P. Roth, Michael A. Calderwood, David E. Hill, Marc Vidal, S Stephen Yi, Nidhi Sahni, Jian Peng, Anne-Claude Gingras, Shantanu Singh, Anne E. Carpenter, Mikko Taipale
- **A principal odor map unifies diverse tasks in human olfactory perception**
  Science (2023)
  Brian K. Lee*, Emily E Mayhew*, Benjamin Sanchez-Lengeling, Jennifer N. Wei, **Wesley W. Qian**, Kelsie Little, Matthew Andres, Britney B. Nguyen, Theresa Moloy, Jacob Yasonik, Jane K. Parker, Richard C. Gerkin, Joel D. Mainland, Alexander B. Wiltschko
- **3D equivariant diffusion for target-aware molecule generation and affinity prediction**
  ICLR (2023)
  Jiaqi Guan*, **Wesley W. Qian***, Xingang Peng, Yufeng Su, Jian Peng, Jianzhu Ma
- **Metabolic activity organizes olfactory representations**
  eLife (2023)
  Wesley W. Qian, Jennifer N. Wei, Benjamin Sanchez-Lengeling, Brian K. Lee, Yunan Luo, Marnix Vlot, Koen Dechering, Jian Peng, Richard C. Gerkin, Alexander B. Wiltschko
- **A central chaperone-like role for 14-3-3 proteins in human cells**
  Molecular Cell (2023)
  Dmitri Segal, Stefan Maier, Giovanni J Mastromarco, **Wesley W. Qian**, Syed Nabeel-Shah, Hyunmin Lee, Gaelen Moore, Jessica Lacoste, Brett Larsen, Zhen-Yuan Lin, Abeeshan Selvabaskaran, Karen Liu, Craig Smibert, Zhaolei Zhang, Jack Greenblatt, Jian Peng, Hyun O Lee, Anne-Claude Gingras, Mikko Taipale
- **Energy-inspired molecular conformation optimization**
  ICLR (2022)
  Jiaqi Guan*, **Wesley W. Qian***, Qiang Liu, Wei-Ying Ma, Jianzhu Ma, Jian Peng

- **ECNet is an evolutionary context-integrated deep learning framework for protein engineering**
  Nature Communications (2021)
  Yunan Luo, Guangde Jiang, Tianhao Yu, Yang Liu, Lam Vo, Hantian Ding, Yufeng Su, **Wesley W. Qian**, Huimin Zhao, Jian Peng
- **Integrating deep neural networks and symbolic inference for organic reactivity prediction**
  ACS National Meeting (2021)
  Wesley W. Qian*, Nathan T. Russell*, Claire L. W. Simons, Yunan Luo, Martin D. Burke, Jian Peng
- **Comprehensive interactome profiling of the human Hsp70 network highlights functional differentiation of J domains**
  Molecular Cell (2021)
  Benjamin L. Piette, Nader Alerasool, Zhen-Yuan Lin, Jessica Lacoste, Mandy Hiu Yi Lam, **Wesley W. Qian**, Stephanie Tran, Brett Larsen, Eric Campos, Jian Peng, Anne-Claude Gingras, Mikko Taipale
- **Evaluating attribution for graph neural networks**
  NeurIPS (2020)
  Benjamin Sanchez-Lengeling, Jennifer Wei, Brian Lee, Emily Reif, Peter Wang, **Wesley W. Qian**, Kevin McCloskey, Lucy Colwell, Alexander B. Wiltschko
- **Batch equalization with a generative adversarial network**
  Bioinformatics (2020)
  Wesley W. Qian, Cassandra Xia, Subhashini Venugopalan, Arunachalam Narayanaswamy, Michelle Dimon, George W. Ashdown, Jake Baum, Jian Peng, D Michael Ando
- **Evolutionary context-integrated deep sequence modeling for protein engineering**
  RECOMB (2020)
  Yunan Luo, Lam Vo, Hantian Ding, Yufeng Su, Yang Liu, **Wesley W. Qian**, Huimin Zhao, Jian Peng

Services

- **Reviewer** for Nature Machine Intelligence (2025), ICML (2024), ICLR (2024), NeurIPS (2023), LoG (2022 & 2023), RECOMB (2021), ISMB (2019 & 2020).
- **Program Committee** for ICML: ML Interpretability for Scientific Discovery Workshop 2020.